Two worlds. The same premises.

A Horos reasoning notebook · Aristotle, *Prior Analytics* I.

This notebook records a modern membership experiment beside an ancient text. Truth in one world is separate from validity across all permitted worlds. The diagrams are modern; the English and teaching Latin are new.

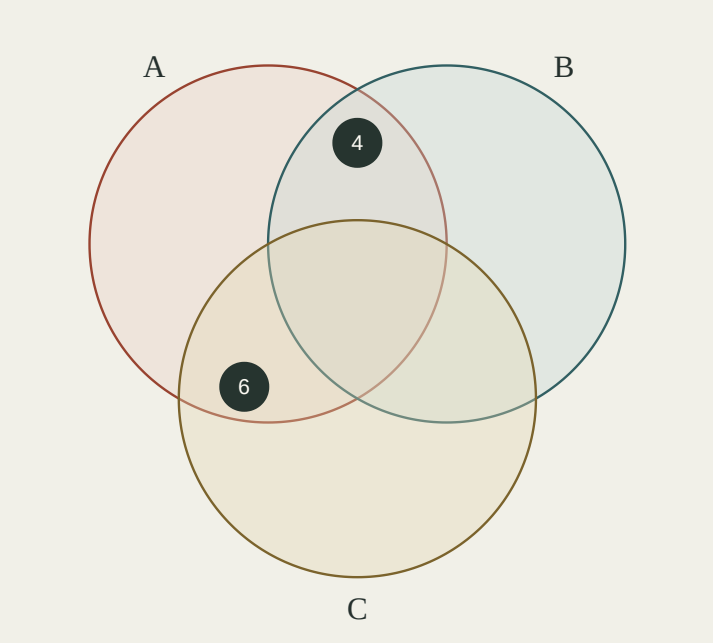
Your explanation

Object 5 is C but not A. Every B is A and no C is B still hold.
Aristotle's horse & stone substitutions preserve the premise forms while changing the conclusion. The two worlds therefore refute the proposed inference, even though its conclusion is true in the first world.

Cited passages: I.4 · 26a2–9; I.1 · 24b18–20.

The two snapshots use the same premises, proposed conclusion, and existence assumptions.

Pinned world: The horse & the stone



Required inhabitants: A, B, C. The universe must be nonempty.

Truth in this world

Statement	Truth	Reason
Premise: Every B is A	True	No object violates this universal.
Premise: No C is B	True	No object violates this universal.
Proposed conclusion: Every C is A	True	No object violates this universal.

World status: agreement.

- Object 4: In A and B; outside C.
- Object 6: In A and C; outside B.

Exhaustive check under these assumptions: 12 permitted patterns; 4 with a true conclusion; 8 counterexamples. Inference: invalid in this modern model.

Where could the counterexample's witness go?

Keep the premises true and make the conclusion false. This asks for **Some C is not A**.

Universals forbid regions; particular statements and existence assumptions require witnesses. The table lists possible object positions, whether or not your current board uses them.

Which regions remain available?

Object position	Condition that forbids it

Object position	Condition that forbids it
1. Outside all three	Available
2. In A; outside B and C	Available
3. In B; outside A and C	Premise 1: Every B is A
4. In A and B; outside C	Available
5. In C; outside A and B	Available
6. In A and C; outside B	Available
7. In B and C; outside A	Premise 1: Every B is A; Premise 2: No C is B
8. In A and B and C	Premise 2: No C is B

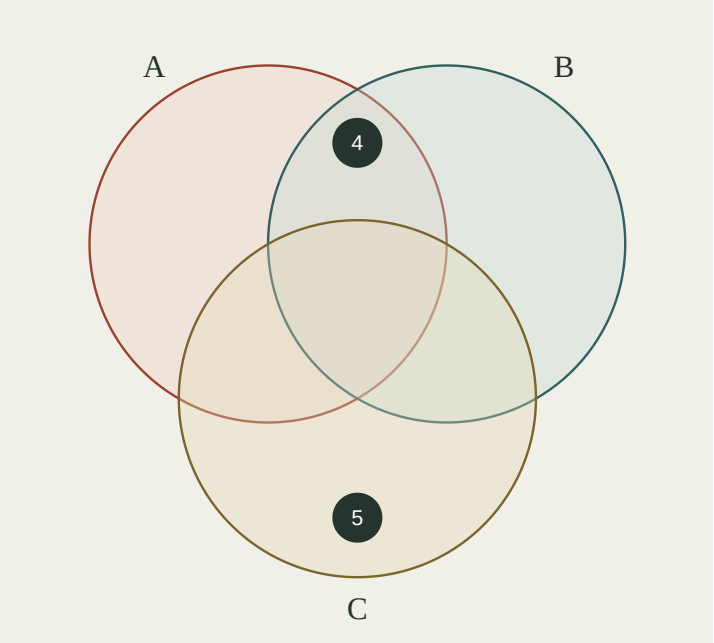
Where each required witness could go

Requirement	Available object positions
The world must contain an object	1, 2, 4, 5, 6
To make the conclusion false: Some C is not A	5
A must have an inhabitant	2, 4, 6
B must have an inhabitant	4
C must have an inhabitant	5, 6

Every required witness has an available region. Put one object in the first available region for each requirement; a single object can serve several requirements. All these witnesses can coexist, so this constructs a counterexample without searching 255 worlds. It need not be the smallest one.

This is a modern explanation using the workshop's membership rules, not a transcription of Aristotle's proof.

Current world: The horse & the stone



Required inhabitants: A, B, C. The universe must be nonempty.

Truth in this world

Statement	Truth	Reason
Premise: Every B is A	True	No object violates this universal.
Premise: No C is B	True	No object violates this universal.
Proposed conclusion: Every C is A	False	Object 5 violates this universal.

World status: counterexample.

- Object 4: In A and B; outside C.
- Object 5: In C; outside A and B.

Exhaustive check under these assumptions: 12 permitted patterns; 4 with a true conclusion; 8 counterexamples. Inference: invalid in this modern model.

Where could the counterexample's witness go?

Keep the premises true and make the conclusion false. This asks for **Some C is not A**.

Universals forbid regions; particular statements and existence assumptions require witnesses. The table lists possible object positions, whether or not your current board uses them.

Which regions remain available?

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4. In A and B; outside C	Available
5. In C; outside A and B	Available
6. In A and C; outside B	Available
7. In B and C; outside A	Premise 1: Every B is A; Premise 2: No C is B
8. In A and B and C	Premise 2: No C is B

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Every required witness has an available region. Put one object in the first available region for each requirement; a single object can serve several requirements. All these witnesses can coexist, so this constructs a counterexample without searching 255 worlds. It need not be the smallest one.

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Why the search is complete

With three classes, an object has one of eight yes/no membership patterns. These propositions ask only which regions are occupied, never how many objects occupy them. Keep one object from each occupied region of any world: all four proposition kinds and each existence condition keep their truth values. Thus every possible counterexample has a representative among the 255 nonempty occupancy patterns. The completely empty pattern is an editable draft but not a permitted universe. This proof applies to the defined model, not to all of Aristotle's logic.

"Every" and "no" are true if their subject class is empty in this model. "Some" and "some not" require a witness; "some" can include all. Requiring an inhabitant is an additional assumption. The existence experiments do not settle Aristotle's historical treatment of empty terms. The source's taking of an instance must be read in its own setting.

Greek A / B / Γ becomes A / B / C. In I.6 Π / P / Σ becomes A / B / C, with N the instance.

What has to follow

I.1 · 24b18–20 · Printed page 24

συλλογισμὸς δέ ἐστι λόγος ἐν ᾧ τεθέντων τινῶν ἕτερόν τι τῶν κειμένων ἐξ ἀνάγκης συμβαίνει τῷ ταῦτα εἶναι.

Close English

A deduction is an account in which, when certain things have been laid down, something different from what was laid down follows of necessity because those things are so.

Teaching Latin · new translation

Syllogismus est ratio in qua, quibusdam positis, aliquid aliud ab iis quae posita sunt necessario sequitur, eo quod haec ita sunt.

The passage distinguishes a consequence that must follow from one that merely happens to be true. ‘Deduction’ translates syllogismos here; this workshop treats only a small, non-modal part of the Prior Analytics. The name Horos comes from ὅρος, ‘term’, defined just before this passage at 24b16–18.

Contextual language notes

Greek	Teaching Latin	Reading note
συλλογισμὸς	syllogismus	syllogismós: a deduction. The word is broader here than a classroom formula of two premises and one conclusion.
τεθέντων τινῶν	quibusdam positis	tetheéntōn tinōn: ‘certain things having been laid down’. A genitive absolute supplies the conditions of the deduction.
ἐξ ἀνάγκης	necessario	ex anánkēs: ‘of necessity’. A true conclusion in one example is weaker than this requirement.
τῷ ταῦτα εἶναι	eo quod haec ita sunt	tôi taûta eînai: ‘because these things are so’. The consequence must depend on what was laid down.

The horse and the stone

I.4 · 26a2–9 · Printed page 26

εἰ δὲ τὸ μὲν πρῶτον παντὶ τῷ μέσῳ ἀκολουθεῖ, τὸ δὲ μέσον μηδενὶ τῷ ἐσχάτῳ ὑπάρχει, οὐκ ἔσται συλλογισμὸς τῶν ἄκρων· οὐδὲν γὰρ ἀναγκαῖον συμβαίνει τῷ ταῦτα εἶναι· καὶ γὰρ παντὶ καὶ μηδενὶ ἐνδέχεται τὸ πρῶτον τῷ ἐσχάτῳ ὑπάρχειν, ὥστε οὔτε τὸ κατὰ μέρος οὔτε τὸ καθόλου γίνεται ἀναγκαῖον· μηδενὸς δὲ ὄντος ἀναγκαίου διὰ τούτων οὐκ ἔσται συλλογισμός. ὅροι τοῦ παντὶ ὑπάρχειν ζῷον—ἄνθρωπος—ἵππος, τοῦ μηδενὶ ζῷον—ἄνθρωπος—λίθος.

Close English

If the first follows every middle, but the middle belongs to none of the last, there will be no deduction connecting the extremes. Nothing necessary follows because these things are so: the first can belong both to every last and to none. Thus neither a particular nor a universal result becomes necessary. With nothing necessary through these premises, there will be no deduction. Terms for belonging to every: animal, human, horse; for belonging to none: animal, human, stone.

Teaching Latin · new abridgment

Si primum omni medio convenit, medium autem nulli extremo, syllogismus extremorum non erit. Nihil enim necessarium ex his sequitur. Termini, ubi primum omni extremo convenit: animal, homo, equus; ubi nulli: animal, homo, lapis.

The Latin condenses the passage. In the two substitutions A = animal, B = human; C is first horse, then stone. The premise forms stay the same while the interpretation of C changes. Our buttons supply small illustrative membership worlds, not Aristotle's actual diagram or a complete inventory of animals. Starting with all three classes inhabited avoids an empty-class shortcut.

Contextual language notes

Greek	Teaching Latin	Reading note
παντὶ τῷ μέσῳ	omni medio	pantì tôi mésōi: ‘to every middle’. The dative names what the predicate applies to. A belongs to every B means every B is A, not every A is B.
μηδενὶ	nulli	mēdenì: ‘to none’. A universal negative excludes every shared object.
τῶν ἄκρων	extremorum	tôn ákrōn: ‘of the extremes’. Here A and C are the extremes; B is the middle term.

Greek	Teaching Latin	Reading note
ὅροι	termini	hóroi: terms. The sequence animal–human–horse names A, B, C; it is not itself a chain of valid implications.
λίθος	lapis	líthos: stone. Substituting this for horse changes the relation of C to A while preserving the two premise forms.

Reversal, with a smaller claim

I.2 · 25a7–9 · Printed page 25

τὴν δὲ κατηγορικὴν ἀντιστρέφειν μὲν ἀναγκαῖον, οὐ μὴν καθόλου ἀλλ᾽ ἐν μέρει, οἷον εἰ πᾶσα ἡδονὴ ἀγαθόν, καὶ ἀγαθόν τι εἶναι ἡδονήν.

Close English

The affirmative must convert, though not universally but in part: for example, if every pleasure is a good, some good is also a pleasure.

Teaching Latin · new translation

Affirmativam converti necesse est, non tamen universaliter sed particulariter: si omnis voluptas bonum est, aliquod bonum voluptas est.

This is a conditional example, not Aristotle endorsing the premise that every pleasure is good. The workshop uses A = good and B = pleasure only as a reading aid. In the modern membership model, the particular conclusion also needs B to have an inhabitant. The checkbox makes that assumption explicit; it does not claim to settle how empty terms function throughout Aristotle's logic. The letter proof is printed nearby at 25a17–19.

Contextual language notes

Greek	Teaching Latin	Reading note
ἀντιστρέφειν	converti	antistréphein: to convert, or reverse the terms. This is a relation between propositions, not turning a word into its opposite.
καθόλου	universaliter	kathólou: universally. Reversal does not preserve the universal affirmative in the proposed form.
ἐν μέρει	particulariter	en mérei: in part, particularly. ‘Some’ needs a witness; it does not mean ‘some but not all’.
ἡδονὴ	voluptas	hēdonḕ: pleasure. Here it belongs to a hypothetical example, not a conclusion about ethics.

Set out one instance

I.6 · 28a23–26 · Printed page 28

ἔστι δὲ καὶ διὰ τοῦ ἀδυνάτου καὶ τῷ ἐκθέσθαι ποιεῖν τὴν ἀπόδειξιν· εἰ γὰρ ἄμφω παντὶ τῷ Σ ὑπάρχει, ἂν ληφθῇ τι τῶν Σ οἷον τὸ Ν, τούτῳ καὶ τὸ Π καὶ τὸ Ρ ὑπάρξει, ὥστε τινὶ τῷ Ρ τὸ Π ὑπάρξει.

Close English

The proof can also be made through the impossible and by setting out an instance. For if both belong to every S, and one of the Ss is taken, say N, both P and R will belong to it; so P will belong to some R.

Teaching Latin · new translation

Probatio etiam per impossibile et per expositionem fieri potest. Si enim utrumque omni S convenit, et sumitur aliquod S, velut N, huic et P et R convenient; itaque P alicui R conveniet.

The printed terms are Π, P, Σ, with N naming the instance. The board renames Π to A, P to B, and Σ to C; object 8 can play the role of N. The Greek describes taking an instance. Allowing C to be empty is a modern contrast, not a claim that Aristotle accepts this counterexample against his own inference. This workshop does not adjudicate the scholarly debate about existential import.

Contextual language notes

Greek	Teaching Latin	Reading note
τῷ ἐκθέσθαι	per expositionem	tôi ekthésthai: ‘by setting out’. Often called ekthesis, the move takes an instance to expose the shared predication.
ἂν ληφθῇ	si sumitur	àn lēphthêi: ‘if it is taken’. The passive subjunctive occurs in a conditional clause. It is the step to inspect when there is no C to take.
τι τῶν Σ	aliquod S	ti tôn S: ‘one of the Ss’. A partitive genitive names the class from which the instance is taken.
τούτῳ	huic	toútōi: ‘to this’. Both predicates apply to the same instance, which supplies the particular conclusion.

The universal chain

I.4 · 25b37–40 · Printed page 25

εἰ γὰρ τὸ Α κατὰ παντὸς τοῦ Β καὶ τὸ Β κατὰ παντὸς τοῦ Γ, ἀνάγκη τὸ Α κατὰ παντὸς τοῦ Γ κατηγορεῖσθαι· πρότερον γὰρ εἴρηται πῶς τὸ κατὰ παντὸς λέγομεν.

Close English

For if A is predicated of every B, and B of every C, A must be predicated of every C; we have already said what we mean by ‘of every’.

Teaching Latin · new translation

Si A de omni B, et B de omni C praedicatur, necesse est A de omni C praedicari; iam enim dictum est quid significet de omni.

The source writes Greek Γ where the board uses C. Take any C. The second premise makes it B, and the first makes it A. If there are no Cs, the modern universal still has no violating instance. A particular conclusion (‘Some C is A’) needs an additional inhabitant. The circles and finite-pattern proof are modern explanations, not diagrams from Bekker's text.

Contextual language notes

Greek	Teaching Latin	Reading note
κατὰ παντὸς	de omni	katà pantòs: ‘of every’. The genitive after katá expresses the subject range of the predication.
κατηγορεῖσθαι	praedicari	katēgoreîsthai: ‘to be predicated’. The predicate A is said of the subject C; English reverses their surface order in ‘Every C is A’.
ἀνάγκη	necesse est	anánkē: necessity; here ‘it is necessary’. This is what a merely agreeing example cannot show.

Edition and text policy

Aristotle, Prior Analytics I. Greek text transcribed from Immanuel Bekker, Aristotelis opera, volume I (Berlin: G. Reimer, 1831), printed pages 24, 25, 26 and 28. Complete page images are retained from Internet Archive item Bekker1831. This is a nineteenth-century printed edition, not an ancient manuscript.

Line breaks and line-end hyphenation are joined; Greek accents and punctuation are normalized for reading. Excerpts have explicit boundaries. English and teaching Latin are new translations for this workshop; Latin is not presented as an ancient or medieval textual witness. The two-worlds Latin is an abridgment. The Greek text and printed scans are public domain.

Source: Internet Archive, Bekker1831 (https://archive.org/details/Bekker1831). Compare Jenkinson's English translation (https://classics.mit.edu/Aristotle/prior.1.i.html).

Bekker 1831 · complete printed page 24

ὅτι οὐκ ἀγαθὸν ἐναντία ἡ τῷ μὴ ἀγαθῷ ὅτι ἀγαθόν· ψευδὴς γὰρ αὕτη. ὥσε καὶ ἡ τῷ ἀγαθῷ ὅτι οὐκ ἀγαθὸν τῇ τῷ ἀγαθῷ ὅτι ἀγαθόν. φανερὸν δὲ ὅτι οὐδὲν διοίσει οὐδ' ἂν καθόλου τιθῶμεν τὴν κατάφασιν· ἡ γὰρ καθόλου ἀπόφασις ἐναντία ἔσται, οἷον τῇ δόξῃ τῇ δοξαζούσῃ ὅτι πᾶν ὃ ἂν ᾖ ἀγαθὸν ἀγαθόν ἐστιν ἡ ὅτι οὐδὲν τῶν ἀγαθῶν ἀγαθόν. ἡ γὰρ τῷ ἀγαθῷ ὅτι ἀγαθόν, εἰ καθόλου τὸ ἀγαθόν, ἡ αὐτή ἐστι τῇ ὅτι ὃ ἂν ᾖ ἀγαθὸν δοξαζούσῃ ὅτι ἀγαθόν· τοῦτο δὲ οὐδὲν διαφέρει τῷ ὅτι πᾶν ὃ ἂν ᾖ ἀγαθὸν ἀγαθόν ἐστιν. ὁμοίως δὲ καὶ ἐπὶ τῷ μὴ ἀγαθῷ. ὥσε εἴπερ ἐπὶ δόξης οὕτως ἔχει, εἰσὶ δὲ αἱ ἐν τῇ φωνῇ καταφάσεις καὶ ἀποφάσεις σύμβολα τῶν ἐν τῇ ψυχῇ, δῆλον ὅτι καὶ καταφάσει ἐναντία μὲν ἀπόφασις ἡ περὶ τοῦ αὐτοῦ καθόλου, οἷον τῇ ὅτι πᾶς ἀγαθὸν ἀγαθὸν ἢ ὅτι πᾶς ἄνθρωπος ἀγαθὸς ἡ ὅτι οὐδεὶς ἢ οὐδείς, ἀντιφατικῶς δὲ ὅτι ἢ οὐ πᾶν ἢ οὐ πᾶς. φανερὸν δὲ ὅτι καὶ ἀληθῆ ἀληθεῖ οὐκ ἐνδέχεται ἐναντίαν εἶναι οὔτε δόξαν οὔτε ἀπόφασιν. ἐναντίαι μὲν γὰρ αἱ περὶ τὰ ἀντικείμενα, περὶ ταῦτα δὲ ἐνδέχεται ἀληθεύειν τὸν αὐτόν· ἅμα δὲ οὐκ ἐνδέχεται τὰ ἐναντία ὑπάρχειν τῷ αὐτῷ.

ΑΝΑΛΥΤΙΚΩΝ ΠΡΟΤΕΡΩΝ Α.

Πρῶτον εἰπεῖν περὶ τί καὶ τίνος ἐστὶν ἡ σκέψις, ὅτι περὶ ἀπόδειξιν καὶ ἐπιστήμης ἀποδεικτικῆς· εἶτα διορίσαι τί ἐστι πρότασις καὶ τί ὅρος καὶ τί συλλογισμός, καὶ ποῖος τέλειος καὶ ποῖος ἀτελής, μετὰ δὲ ταῦτα τί τὸ ἐν ὅλῳ εἶναι ἢ μὴ εἶναι τόδε τῷδε, καὶ τί λέγομεν τὸ κατὰ παντὸς ἢ μηδενὸς κατηγορεῖσθαι.

Πρότασις μὲν οὖν ἐστὶ λόγος καταφατικὸς ἢ ἀποφατικὸς τινὸς κατά τινος. ὅτος δὲ ἢ καθόλου ἢ ἐν μέρει ἢ ἀδιόριστος. λέγω δὲ καθόλου μὲν τὸ παντὶ ἢ μηδενὶ ὑπάρχειν, ἐν μέρει δὲ τὸ τινὶ ἢ μὴ τινὶ ἢ μὴ παντὶ ὑπάρχειν, ἀδιόριστον δὲ τὸ ὑπάρχειν ἢ μὴ ὑπάρχειν ἄνευ τοῦ καθόλου, ἢ κατὰ μέρος, οἷον τὸ τῶν ἐναντίων εἶναι τὴν αὐτὴν ἐπιστήμην ἢ τὸ τὴν ἡδονὴν μὴ εἶναι ἀγαθόν. διαφέρει δὲ ἡ ἀποδεικτικὴ πρότασις τῆς διαλεκτικῆς, ὅτι ἡ μὲν ἀποδεικτικὴ λῆψις θατέρου μορίου τῆς ἀντιφάσεώς ἐστιν (ὁ γὰρ ἐρωτᾷ ἀλλὰ λαμβάνει ὁ ἀποδεικνύων), ἡ δὲ διαλεκτικὴ ἐρώτησις ἀντιφάσεώς ἐστιν. ὁθὲν δὲ διοίσει πρὸς τὸ γενέσθαι τὸν ἑκατέρου συλλογισμόν· καὶ γὰρ ὁ ἀποδεικνύων καὶ ὁ ἐρωτῶν συλλογίζεται λαβών τι κατά τινος ὑπάρχειν ἢ μὴ ὑπάρχειν. ὥσε ἔσται συλλογιστικὴ μὲν πρότασις ἁπλῶς κατάφασις ἢ ἀπόφασίς τινος κατά τινος τὸν εἰρημένον τρόπον, ἀποδεικτικὴ δέ, ἐὰν ἀληθὴς ᾖ καὶ διὰ τῶν ἐξ ἀρχῆς ὑποθέσεων εἰλημμένη, διαλεκτικὴ δὲ πυνθανομένῳ μὲν ἐρώτησις ἀντιφάσεως, συλλογιζομένῳ δὲ λῆψις τοῦ φαινομένου καὶ ἐνδόξου, καθάπερ ἐν τοῖς Τοπικοῖς εἴρηται. τί μὲν οὖν ἐστὶ πρότασις, καὶ τί διαφέρει συλλογιστικὴ καὶ ἀποδεικτικὴ καὶ διαλεκτική, δι' ἀκριβείας μὲν ἐν τοῖς ἑπομένοις ῥηθήσεται, πρὸς δὲ τὴν παροῦσαν χρείαν ἱκανῶς ἡμῖν διωρίσθω τὰ νῦν. ὅρον δὲ καλῶ εἰς ὃν διαλύεται ἡ πρότασις, οἷον τό τε κατηγορούμενον καὶ τὸ καθ' οὗ κατηγορεῖται, ἢ προσιθεμένης ἢ διαιρουμένης τῆς εἶναι καὶ μὴ εἶναι. συλλογισμὸς δέ ἐστι λόγος ἐν ᾧ τεθέντων τινῶν ἕτερόν τι τῶν κειμένων ἐξ ἀνάγκης συμβαίνει τῷ ταῦτα εἶναι. λέγω δὲ τῷ ταῦτα εἶναι τὸ διὰ ταῦτα συμβαίνειν, τὸ δὲ διὰ ταῦτα συμβαίνειν τὸ μηδενὸς ἔξωθεν ὅρου προσδεῖν πρὸς τὸ γενέσθαι τὸ ἀναγκαῖον. τέλειον μὲν οὖν καλῶ συλλογισμὸν τὸν μηδενὸς ἄλλου προσδεόμενον παρὰ τὰ εἰλημμένα πρὸς τὸ φανῆναι τὸ ἀναγκαῖον, ἀτελῆ δὲ τὸν προσδεόμενον ἢ ἑνὸς ἢ πλειόνων, ἃ ἔστι μὲν ἀναγκαῖα διὰ τῶν ὑποκειμένων ὅρων, ἃ μὴν εἴληπται διὰ προτάσεων. τὸ δὲ ἐν ὅλῳ εἶναι ἕτερον ἑτέρῳ καὶ τὸ κατὰ παντὸς κατηγορεῖσθαι θατέρῳ θατέρου ταὐτόν ἐστιν. λέγομεν δὲ τὸ κατὰ παντὸς κατηγορεῖσθαι, ὅταν μηδὲν ᾖ λαβεῖν τῶν τοῦ ὑποκειμένου καθ' οὗ θατέρου ὁ λεχθήσεται· καὶ τὸ κατὰ μηδενὸς ὡσαύτως.

1. ὅτι—ὅτι ἀγαθόν rc *A* supra lituram ‖ ἀληθὴς *A*, ἀληθῆς ψευδὴς *B*. ‖ 6. ἡ—ὅτι ἀγαθόν om *A*. ‖ 7. τῇ πᾶν ὅτι *A* et pr *B*. ‖ ὃ om *C*. ‖ 9. δὲ om. *A*.

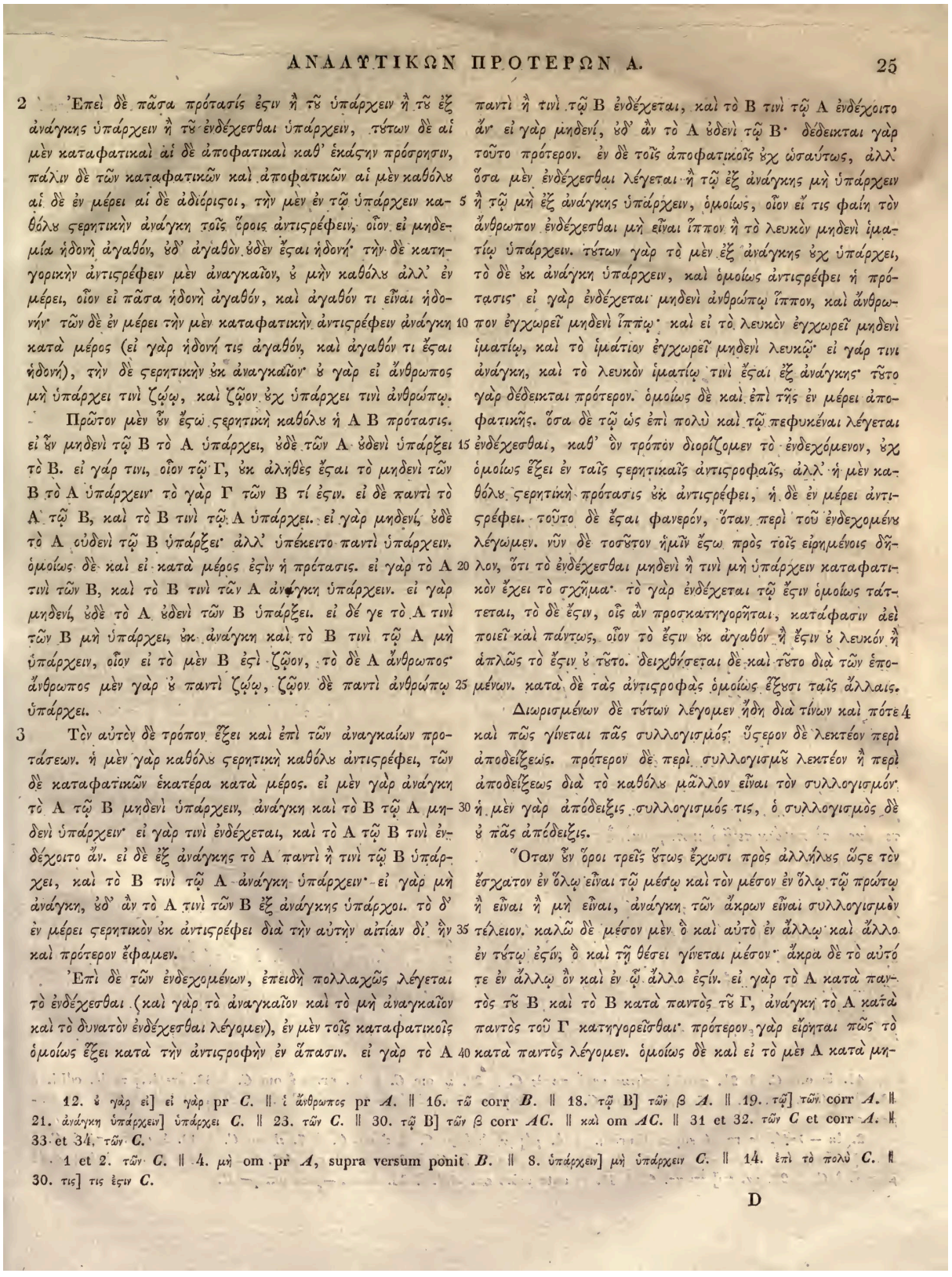
4. τῇ om *BC* et pr *A*. ‖ 5. ἀντιφατικῶς *A*. ‖ ὅτι ἢ—πᾶς] ἡ ὁ πᾶν καὶ ἡ ὁ πᾶς *C*. ‖ 6. ἀληθεῖ ἀληθῆ *C*.

29. ἡ ἀπόφασίς om pr *C*.

18. καὶ] ἢ *C*. ‖ 19. τι—ἀνάγκης rc *A* supra lituram. ‖ τῶν κειμένων om *B*. ‖ 20. τῷ *C*. ‖ 21. τῷ δὲ *ABC*. ‖ 26. ἐν—29. ᾖ rc *A* supra lituram. ‖ 27. ἕτερον ἐν ἑτέρῳ *BC*. ‖ 29. τῶν om *BC*.

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2 Ἐπεὶ δὲ πᾶσα πρότασίς ἐστιν ἢ τοῦ ὑπάρχειν ἢ τοῦ ἐξ ἀνάγκης ὑπάρχειν ἢ τοῦ ἐνδέχεσθαι ὑπάρχειν, τούτων δὲ αἱ μὲν καταφατικαὶ αἱ δὲ ἀποφατικαὶ καθ' ἑκάστην πρόσρησιν, πάλιν δὲ τῶν καταφατικῶν καὶ ἀποφατικῶν αἱ μὲν καθόλου αἱ δὲ ἐν μέρει αἱ δὲ ἀδιόριστοι, τὴν μὲν ἐν τῷ ὑπάρχειν καθόλου στερητικὴν ἀνάγκη τοῖς ὅροις ἀντιστρέφειν, οἷον εἰ μηδεμία ἡδονὴ ἀγαθόν, ὁδ' ἀγαθὸν ὐδὲν ἔσαι ἡδονή· τὴν δὲ κατηγορικὴν ἀντιστρέφειν μὲν ἀναγκαῖον, ὐ μὴν καθόλου ἀλλ' ἐν μέρει, οἷον εἰ πᾶσα ἡδονὴ ἀγαθόν, καὶ ἀγαθόν τι εἶναι ἡδονήν· τῶν δὲ ἐν μέρει τὴν μὲν καταφατικὴν ἀντιστρέφειν ἀνάγκη κατὰ μέρος (εἰ γὰρ ἡδονή τις ἀγαθόν, καὶ ἀγαθόν τι ἔσαι ἡδονή), τὴν δὲ στερητικὴν ὐκ ἀναγκαῖον· ὐ γὰρ εἰ ἄνθρωπος μὴ ὑπάρχει τινὶ ζῴῳ, καὶ ζῷον ὐχ ὑπάρχει τινὶ ἀνθρώπῳ.

Πρῶτον μὲν ὧν ἔσω στερητικὴ καθόλου ἡ Α Β πρότασις. εἰ ὧν μηδενὶ τῷ Β τὸ Α ὑπάρχει, ὐδὲ τῶν Α ὐδενὶ ὑπάρξει τὸ Β. εἰ γάρ τινι, οἷον τῷ Γ, ὐκ ἀληθὲς ἔσαι τὸ μηδενὶ τῶν Β τὸ Α ὑπάρχειν· τὸ γὰρ Γ τῶν Β τί ἐστιν. εἰ δὲ παντὶ τὸ Α τῷ Β, καὶ τὸ Β τινὶ τῷ Α ὑπάρχει. εἰ γὰρ μηδενί, ὐδὲ τὸ Α οὐδενὶ τῷ Β ὑπάρξει· ἀλλ' ὑπέκειτο παντὶ ὑπάρχειν. ὁμοίως δὲ καὶ εἰ κατὰ μέρος ἐστὶν ἡ πρότασις. εἰ γὰρ τὸ Α τινὶ τῶν Β, καὶ τὸ Β τινὶ τῶν Α ἀνάγκη ὑπάρχειν. εἰ γὰρ μηδενί, ὐδὲ τὸ Α ὐδενὶ τῶν Β ὑπάρξει. εἰ δέ γε τὸ Α τινὶ τῶν Β μὴ ὑπάρχει, ὐκ ἀνάγκη καὶ τὸ Β τινὶ τῷ Α μὴ ὑπάρχειν, οἷον εἰ τὸ μὲν Β ἐστὶ ζῷον, τὸ δὲ Α ἄνθρωπος· ἄνθρωπος μὲν γὰρ ὐ παντὶ ζῴῳ, ζῷον δὲ παντὶ ἀνθρώπῳ ὑπάρχει.

3 Τὸν αὐτὸν δὲ τρόπον ἔζει καὶ ἐπὶ τῶν ἀναγκαίων προτάσεων. ἡ μὲν γὰρ καθόλου στερητικὴ καθόλου ἀντιστρέφει, τῶν δὲ καταφατικῶν ἑκατέρα κατὰ μέρος. εἰ μὲν γὰρ ἀνάγκη τὸ Α τῷ Β μηδενὶ ὑπάρχειν, ἀνάγκη καὶ τὸ Β τῷ Α μηδενὶ ὑπάρχειν· εἰ γὰρ τινὶ ἐνδέχεται, καὶ τὸ Α τῷ Β τινὶ ἐνδέχοιτο ἄν. εἰ δὲ ἐξ ἀνάγκης τὸ Α παντὶ ἢ τινὶ τῷ Β ὑπάρχει, καὶ τὸ Β τινὶ τῷ Α ἀνάγκη ὑπάρχειν· εἰ γὰρ μὴ ἀνάγκη, ὁδ' ἂν τὸ Α τινὶ τῶν Β ἐξ ἀνάγκης ὑπάρχοι. τὸ δ' ἐν μέρει στερητικὸν ὐκ ἀντιστρέφει διὰ τὴν αὐτὴν αἰτίαν δι' ἣν καὶ πρότερον ἔφαμεν.

Ἐπὶ δὲ τῶν ἐνδεχομένων, ἐπειδὴ πολλαχῶς λέγεται τὸ ἐνδέχεσθαι (καὶ γὰρ τὸ ἀναγκαῖον καὶ τὸ μὴ ἀναγκαῖον καὶ τὸ δυνατὸν ἐνδέχεσθαι λέγομεν), ἐν μὲν τοῖς καταφατικοῖς ὁμοίως ἔζει κατὰ τὴν ἀντιστροφὴν ἐν ἅπασιν. εἰ γὰρ τὸ Α παντὶ ἢ τινὶ τῷ Β ἐνδέχεται, καὶ τὸ Β τινὶ τῷ Α ἐνδέχοιτο ἄν· εἰ γὰρ μηδενί, ὐδ' ἂν τὸ Α ὐδενὶ τῷ Β· δέδεικται γὰρ τοῦτο πρότερον. ἐν δὲ τοῖς ἀποφατικοῖς ὐχ ὡσαύτως, ἀλλ' ὅσα μὲν ἐνδέχεσθαι λέγεται ἢ τῷ ἐξ ἀνάγκης μὴ ὑπάρχειν ἢ τῷ μὴ ἐξ ἀνάγκης ὑπάρχειν, ὁμοίως, οἷον εἴ τις φαίη τὸν ἄνθρωπον ἐνδέχεσθαι μὴ εἶναι ἵππον ἢ τὸ λευκὸν μηδενὶ ἱματίῳ ὑπάρχειν. τούτων γὰρ τὸ μὲν ἐξ ἀνάγκης ὐχ ὑπάρχει, τὸ δὲ ὐκ ἀνάγκη ὑπάρχειν, καὶ ὁμοίως ἀντιστρέφει ἡ πρότασις· εἰ γὰρ ἐνδέχεται μηδενὶ ἀνθρώπῳ ἵππον, καὶ ἄνθρωπον ἐγχωρεῖ μηδενὶ ἵππῳ· καὶ εἰ τὸ λευκὸν ἐγχωρεῖ μηδενὶ ἱματίῳ, καὶ τὸ ἱμάτιον ἐγχωρεῖ μηδενὶ λευκῷ· εἰ γάρ τινι ἀνάγκη, καὶ τὸ λευκὸν ἱματίῳ τινὶ ἔσαι ἐξ ἀνάγκης· τοῦτο γὰρ δέδεικται πρότερον. ὁμοίως δὲ καὶ ἐπὶ τῆς ἐν μέρει ἀποφατικῆς. ὅσα δὲ τῷ ὡς ἐπὶ πολὺ καὶ τῷ πεφυκέναι λέγεται ἐνδέχεσθαι, καθ' ὃν τρόπον διορίζομεν τὸ ἐνδεχόμενον, ὐχ ὁμοίως ἔζει ἐν ταῖς στερητικαῖς ἀντιστροφαῖς, ἀλλ' ἡ μὲν καθόλου στερητικὴ πρότασις ὐκ ἀντιστρέφει, ἡ δὲ ἐν μέρει ἀντιστρέφει. τοῦτο δὲ ἔσαι φανερόν, ὅταν περὶ τοῦ ἐνδεχομένου λέγωμεν. νῦν δὲ τοσοῦτον ἡμῖν ἔσω πρὸς τοῖς εἰρημένοις δῆλον, ὅτι τὸ ἐνδέχεσθαι μηδενὶ ἢ τινὶ μὴ ὑπάρχειν καταφατικὸν ἔχει τὸ σχῆμα· τὸ γὰρ ἐνδέχεται τῷ ἔστιν ὁμοίως τάττεται, τὸ δὲ ἔστιν, οἷς ἂν προσκατηγορῆται, κατάφασιν ἀεὶ ποιεῖ καὶ πάντως, οἷον τὸ ἔστιν ὐκ ἀγαθόν ἢ ἔστιν ὐ λευκόν ἢ ἁπλῶς τὸ ἔστιν ὐ τοῦτο. δειχθήσεται δὲ καὶ τοῦτο διὰ τῶν ἑπομένων. κατὰ δὲ τὰς ἀντιστροφὰς ὁμοίως ἕξουσι ταῖς ἄλλαις.

4 Διωρισμένων δὲ τούτων λέγομεν ἤδη διά τίνων καὶ πότε καὶ πῶς γίνεται πᾶς συλλογισμός· ὕστερον δὲ λεκτέον περὶ ἀποδείξεως. πρότερον δὲ περὶ συλλογισμοῦ λεκτέον ἢ περὶ ἀποδείξεως διὰ τὸ καθόλου μᾶλλον εἶναι τὸν συλλογισμόν· ἡ μὲν γὰρ ἀπόδειξις συλλογισμός τις, ὁ συλλογισμὸς δὲ ὐ πᾶς ἀπόδειξις.

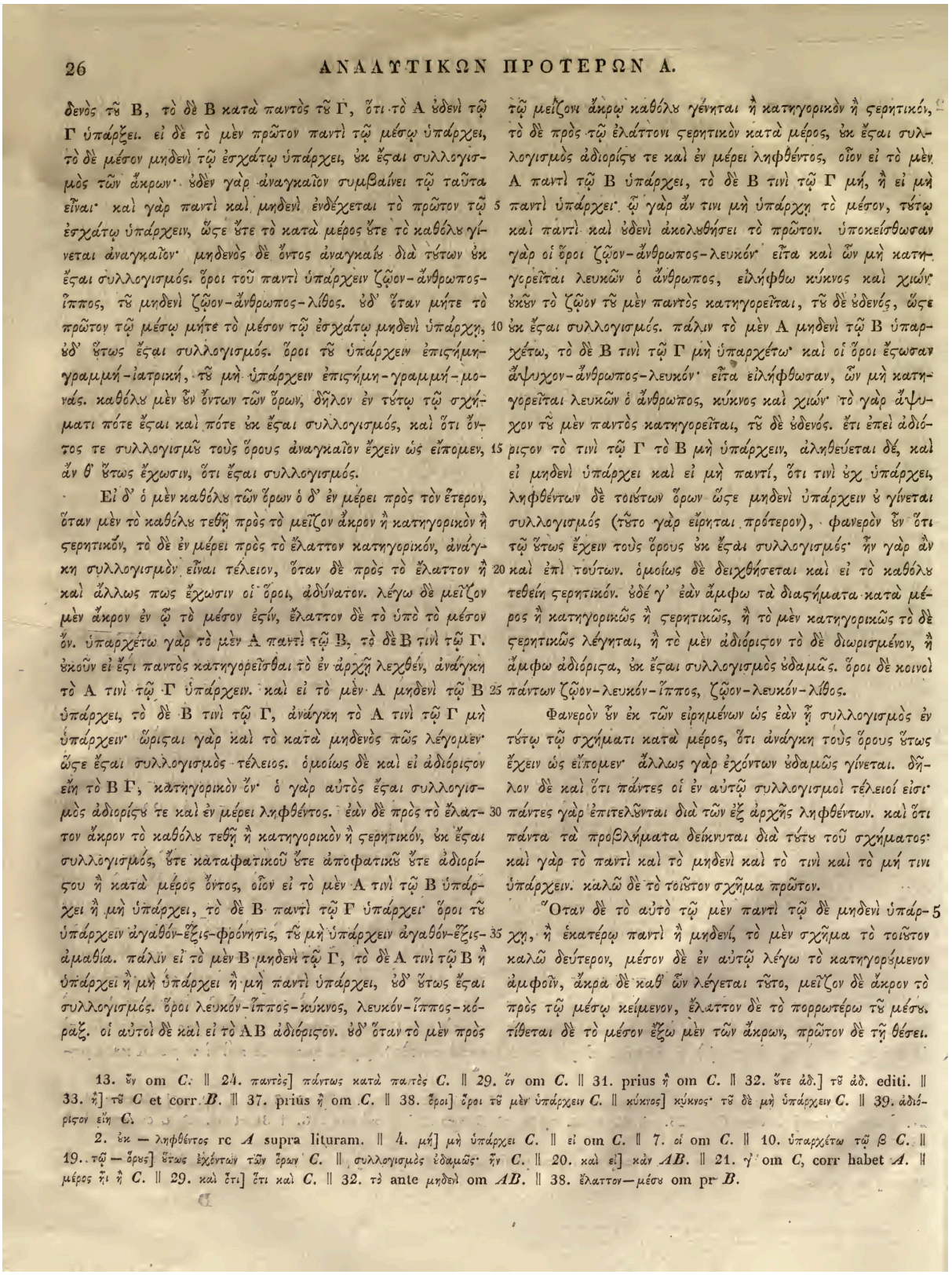
Ὅταν οὖν ὅροι τρεῖς οὕτως ἔχωσι πρὸς ἀλλήλους ὥστε τὸν ἔσχατον ἐν ὅλῳ εἶναι τῷ μέσῳ καὶ τὸν μέσον ἐν ὅλῳ τῷ πρώτῳ ἢ εἶναι ἢ μὴ εἶναι, ἀνάγκη τῶν ἄκρων εἶναι συλλογισμὸν τέλειον. καλῶ δὲ μέσον μὲν ὃ καὶ αὐτὸ ἐν ἄλλῳ καὶ ἄλλο ἐν τούτῳ ἐστίν, ὃ καὶ τῇ θέσει γίνεται μέσον· ἄκρα δὲ τό τε αὐτὸ ἐν ἄλλῳ ὂν καὶ ἐν ᾧ ἄλλο ἐστίν. εἰ γὰρ τὸ Α κατὰ παντὸς τοῦ Β καὶ τὸ Β κατὰ παντὸς τοῦ Γ, ἀνάγκη τὸ Α κατὰ παντὸς τοῦ Γ κατηγορεῖσθαι· πρότερον γὰρ εἴρηται πῶς τὸ κατὰ παντὸς λέγομεν. ὁμοίως δὲ καὶ εἰ τὸ μὲν Α κατὰ μη-

12. ὐ γὰρ εἰ] εἰ γὰρ pr *C.* ‖ ἰ ἄνθρωπος pr *A.* ‖ 16. τῷ corr *B.* ‖ 18. τῷ Β] τῶν β *A.* ‖ 19. τῷ] τῶν corr *A.* ‖ 21. ἀνάγκη ὑπάρχειν] ὑπάρχει *C.* ‖ 23. τῶν *C.* ‖ 30. τῷ Β] τῶν β corr *AC.* ‖ καὶ om *AC.* ‖ 31 et 32. τῶν *C* et corr *A.* ‖ 33 et 34. τῶν *C.*

1 et 2. τῶν *C.* ‖ 4. μὴ om pr *A*, supra versum ponit *B.* ‖ 8. ὑπάρχειν] μὴ ὑπάρχειν *C.* ‖ 14. ἐπὶ τὸ πολὺ *C.* ‖ 30. τις] τις ἔστιν *C.*

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δενὸς τῷ Β, τὸ δὲ Β κατὰ παντὸς τῷ Γ, ὅτι τὸ Α ὐδενὶ τῷ Γ ὑπάρξει. εἰ δὲ τὸ μὲν πρῶτον παντὶ τῷ μέσῳ ὑπάρχει, τὸ δὲ μέσον μηδενὶ τῷ ἐσχάτῳ ὑπάρχει, οὐκ ἔσαι συλλογισμὸς τῶν ἄκρων· οὐδὲν γὰρ ἀναγκαῖον συμβαίνει τῷ ταῦτα εἶναι· καὶ γὰρ παντὶ καὶ μηδενὶ ἐνδέχεται τὸ πρῶτον τῷ ἐσχάτῳ ὑπάρχειν, ὥστε ὅτε τὸ κατὰ μέρος ὅτε τὸ καθόλου γίνεται ἀναγκαῖον· μηδενὸς δὲ ὄντος ἀναγκαῖα διὰ τούτων οὐκ ἔσαι συλλογισμός. ὅροι τοῦ παντὶ ὑπάρχειν ζῷον-ἄνθρωπος-ἵππος, τῷ μηδενὶ ζῷον-ἄνθρωπος-λίθος. οὐδ' ὅταν μήτε τὸ πρῶτον τῷ μέσῳ μήτε τὸ μέσον τῷ ἐσχάτῳ μηδενὶ ὑπάρχῃ, οὐδ' ὅτως ἔσαι συλλογισμός. ὅροι τῷ ὑπάρχειν ἐπιστήμη-γραμμή-ἰατρική, τῷ μὴ ὑπάρχειν ἐπιστήμη-γραμμή-μονάς. καθόλου μὲν οὖν ὄντων τῶν ὅρων, δῆλον ἐν τούτῳ τῷ σχήματι πότε ἔσαι καὶ πότε οὐκ ἔσαι συλλογισμός, καὶ ὅτι ὄντος τε συλλογισμοῦ τοὺς ὅρους ἀναγκαῖον ἔχειν ὡς εἴπομεν, ἂν θ' ὅτως ἔχωσιν, ὅτι ἔσαι συλλογισμός.

Εἰ δ' ὁ μὲν καθόλου τῶν ὅρων ὁ δ' ἐν μέρει πρὸς τὸν ἕτερον, ὅταν μὲν τὸ καθόλου τεθῇ πρὸς τὸ μεῖζον ἄκρον ἢ κατηγορικὸν ἢ στερητικόν, τὸ δὲ ἐν μέρει πρὸς τὸ ἔλαττον κατηγορικόν, ἀνάγκη συλλογισμὸν εἶναι τέλειον, ὅταν δὲ πρὸς τὸ ἔλαττον ἢ καὶ ἄλλως πως ἔχωσιν οἱ ὅροι, ἀδύνατον. λέγω δὲ μεῖζον μὲν ἄκρον ἐν ᾧ τὸ μέσον ἐσίν, ἔλαττον δὲ τὸ ὑπὸ τὸ μέσον ὄν. ὑπαρχέτω γὰρ τὸ μὲν Α παντὶ τῷ Β, τὸ δὲ Β τινὶ τῷ Γ. ἀκοῦν εἰ ἔσι παντὸς κατηγορεῖσθαι τὸ ἐν ἀρχῇ λεχθέν, ἀνάγκη τὸ Α τινὶ τῷ Γ ὑπάρχειν. καὶ εἰ τὸ μὲν Α μηδενὶ τῷ Β ὑπάρχει, τὸ δὲ Β τινὶ τῷ Γ, ἀνάγκη τὸ Α τινὶ τῷ Γ μὴ ὑπάρχειν· ὥρισαι γὰρ καὶ τὸ κατὰ μηδενὸς πῶς λέγομεν· ὥστε ἔσαι συλλογισμὸς τέλειος. ὁμοίως δὲ καὶ εἰ ἀδιόριστον εἴη τὸ Β Γ, κατηγορικὸν ὄν· ὁ γὰρ αὐτὸς ἔσαι συλλογισμὸς ἀδιορίστε τε καὶ ἐν μέρει ληφθέντος. ἐὰν δὲ πρὸς τὸ ἔλαττον ἄκρον τὸ καθόλου τεθῇ ἢ κατηγορικὸν ἢ στερητικόν, οὐκ ἔσαι συλλογισμός, ὅτε καταφατικοῦ ὅτε ἀποφατικοῦ ὅτε ἀδιορίσου ἢ κατὰ μέρος ὄντος, οἷον εἰ τὸ μὲν Α τινὶ τῷ Β ὑπάρχει ἢ μὴ ὑπάρχει, τὸ δὲ Β παντὶ τῷ Γ ὑπάρχει· ὅροι τῷ ὑπάρχειν ἀγαθόν-ἕξις-φρόνησις, τῷ μὴ ὑπάρχειν ἀγαθόν-ἕξις-ἀμαθία. πάλιν εἰ τὸ μὲν Β μηδενὶ τῷ Γ, τὸ δὲ Α τινὶ τῷ Β ἢ ὑπάρχει ἢ μὴ ὑπάρχει ἢ μὴ παντὶ ὑπάρχει, οὐδ' ὅτως ἔσαι συλλογισμός. ὅροι λευκόν-ἵππος-κύκνος, λευκόν-ἵππος-κόραξ. οἱ αὐτοὶ δὲ καὶ εἰ τὸ ΑΒ ἀδιόριστον. οὐδ' ὅταν τὸ μὲν πρὸς τῷ μείζονι ἄκρῳ καθόλου γένηται ἢ κατηγορικὸν ἢ στερητικόν, τὸ δὲ πρὸς τῷ ἐλάττονι στερητικὸν κατὰ μέρος, οὐκ ἔσαι συλλογισμὸς ἀδιορίστε τε καὶ ἐν μέρει ληφθέντος, οἷον εἰ τὸ μὲν Α παντὶ τῷ Β ὑπάρχει, τὸ δὲ Β τινὶ τῷ Γ μή, ἢ εἰ μὴ παντὶ ὑπάρχει. ᾧ γὰρ ἄν τινι μὴ ὑπάρχῃ τὸ μέσον, τούτῳ καὶ παντὶ καὶ οὐδενὶ ἀκολουθήσει τὸ πρῶτον. ὑποκείσθωσαν γὰρ οἱ ὅροι ζῷον-ἄνθρωπος-λευκόν· εἶτα καὶ ὧν μὴ κατηγορεῖται λευκῶν ὁ ἄνθρωπος, εἰλήφθω κύκνος καὶ χιών· ἐκεῖν τὸ ζῷον τῷ μὲν παντὸς κατηγορεῖται, τῷ δὲ οὐδενός, ὥστε οὐκ ἔσαι συλλογισμός. πάλιν τὸ μὲν Α μηδενὶ τῷ Β ὑπαρχέτω, τὸ δὲ Β τινὶ τῷ Γ μὴ ὑπαρχέτω· καὶ οἱ ὅροι ἔσωσαν ἄψυχον-ἄνθρωπος-λευκόν· εἶτα εἰλήφθωσαν, ὧν μὴ κατηγορεῖται λευκῶν ὁ ἄνθρωπος, κύκνος καὶ χιών· τὸ γὰρ ἄψυχον τῷ μὲν παντὸς κατηγορεῖται, τῷ δὲ οὐδενός. ἔτι ἐπεὶ ἀδιόριζον τὸ τινὶ τῷ Γ τὸ Β μὴ ὑπάρχειν, ἀληθεύεται δέ, καὶ εἰ μηδενὶ ὑπάρχει καὶ εἰ μὴ παντί, ὅτι τινὶ οὐχ ὑπάρχει, ληφθέντων δὲ τοιούτων ὅρων ὥσε μηδενὶ ὑπάρχειν ὑ γίνεται συλλογισμός (τοῦτο γὰρ εἴρηται πρότερον), φανερὸν ἦν ὅτι τῷ ὅτως ἔχειν τοὺς ὅρους οὐκ ἔσαι συλλογισμός· ἦν γὰρ ἂν καὶ ἐπὶ τούτων. ὁμοίως δὲ δειχθήσεται καὶ εἰ τὸ καθόλου τεθείη στερητικόν. οὐδέ γ' ἐὰν ἄμφω τὰ διαστήματα κατὰ μέρος ἢ κατηγορικῶς ἢ στερητικῶς, ἢ τὸ μὲν κατηγορικῶς τὸ δὲ στερητικῶς λέγηται, ἢ τὸ μὲν ἀδιόριστον τὸ δὲ διωρισμένον, ἢ ἄμφω ἀδιόριστα, οὐκ ἔσαι συλλογισμὸς οὐδαμῶς. ὅροι δὲ κοινοὶ πάντων ζῷον-λευκόν-ἵππος, ζῷον-λευκόν-λίθος.

Φανερὸν οὖν ἐκ τῶν εἰρημένων ὡς ἐὰν ᾖ συλλογισμὸς ἐν τούτῳ τῷ σχήματι κατὰ μέρος, ὅτι ἀνάγκη τοὺς ὅρους ὅτως ἔχειν ὡς εἴπομεν· ἄλλως γὰρ ἐχόντων οὐδαμῶς γίνεται. δῆλον δὲ καὶ ὅτι πάντες οἱ ἐν αὐτῷ συλλογισμοὶ τέλειοί εἰσι· πάντες γὰρ ἐπιτελοῦνται διὰ τῶν ἐξ ἀρχῆς ληφθέντων. καὶ ὅτι πάντα τὰ προβλήματα δείκνυται διὰ τούτου τοῦ σχήματος· καὶ γὰρ τὸ παντὶ καὶ τὸ μηδενὶ καὶ τὸ τινὶ καὶ τὸ μή τινι ὑπάρχειν. καλῶς δὲ τὸ τοιοῦτον σχῆμα πρῶτον.

Ὅταν δὲ τὸ αὐτὸ τῷ μὲν παντὶ τῷ δὲ μηδενὶ ὑπάρχῃ, ἢ ἑκατέρῳ παντὶ ἢ μηδενί, τὸ μὲν σχῆμα τὸ τοιοῦτον καλῶ δεύτερον, μέσον δὲ ἐν αὐτῷ λέγω τὸ κατηγορούμενον ἀμφοῖν, ἄκρα δὲ καθ' ὧν λέγεται τοῦτο, μεῖζον δὲ ἄκρον τὸ πρὸς τῷ μέσῳ κείμενον, ἔλαττον δὲ τὸ πορρωτέρω τοῦ μέσου. τίθεται δὲ τὸ μέσον ἔξω μὲν τῶν ἄκρων, πρῶτον δὲ τῇ θέσει.

13. ἦν om *C*. ‖ 24. παντὸς] πάντως κατὰ παντὸς *C*. ‖ 29. ἔν om *C*. ‖ 31. prius ἢ om *C*. ‖ 32. ὅτε ἀδ.] τοῦ ἀδ. editi. ‖ 33. ἢ] τῷ *C* et corr *B*. ‖ 37. prius ἢ om *C*. ‖ 38. ὅροι] ὅροι τῷ μὲν ὑπάρχειν *C*. ‖ κύκνος] κύκνος· τῷ δὲ μὴ ὑπάρχειν *C*. ‖ 39. ἀδιόριστον εἴη *C*.

2. οὐκ — ληφθέντος rc *A* supra lituram. ‖ 4. μή] μὴ ὑπάρχει *C*. ‖ εἰ om *C*. ‖ 7. οἱ om *C*. ‖ 10. ὑπαρχέτω τῷ β *C*. ‖ 19. τῷ — ὅρους] ὅτως ἐκέντων τῶν ὅρων *C*. ‖ συλλογισμὸς ὐδαμῶς· ἦν *C*. ‖ 20. καὶ εἰ] κἀν *AB*. ‖ 21. γ' om *C*, corr habet *A*. ‖ μέρος ἢ ἢ *C*. ‖ 29. καὶ ὅτι] ὅτι καὶ *C*. ‖ 32. τὸ ante μηδενὶ om *AB*. ‖ 38. ἔλαττον—μέσου om pr *B*.

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Φανερὸν ἓν ἐκ τῶν εἰρημένων ὅτι ἐάν τε ὅτως ἔχωσιν οἱ ὅροι πρὸς ἀλλήλους ὡς ἐλέχθη, γίνεται συλλογισμὸς ἐξ ἀνάγκης, ἄν τ' ᾖ συλλογισμός, ἀνάγκη τοὺς ὅρους ὅτως ἔχειν. δῆλον δὲ καὶ ὅτι πάντες ἀτελεῖς εἰσὶν οἱ ἐν τούτῳ τῷ σχήματι συλλογισμοί (πάντες γὰρ ἐπιτελοῦνται προσλαμβανομένων τινῶν, ἃ ἢ ἐνυπάρχει τοῖς ὅροις ἐξ ἀνάγκης ἢ τίθενται ὡς ὑποθέσεις, οἷον ὅταν διὰ τῆς ἀδυνάτου δεικνύωμεν), καὶ ὅτι ὁ γίνεται καταφατικὸς συλλογισμὸς διὰ τούτου τοῦ σχήματος, ἀλλὰ πάντες ζερητικοί, καὶ οἱ καθόλου καὶ οἱ κατὰ μέρος.

6 Ἐὰν δὲ τῷ αὐτῷ τὸ μὲν παντὶ τὸ δὲ μηδενὶ ὑπάρχῃ, ἢ ἄμφω παντὶ ἢ μηδενί, τὸ μὲν σχῆμα τὸ τοιοῦτον καλῶ τρίτον, μέσον δ' ἐν αὐτῷ λέγω καθ' οὗ ἄμφω τὰ κατηγορούμενα, ἄκρα δὲ τὰ κατηγορούμενα, μεῖζον δ' ἄκρον τὸ πορρώτερον τοῦ μέσου, ἔλαττον δὲ τὸ ἐγγύτερον. τίθεται δὲ τὸ μέσον ἔξω μὲν τῶν ἄκρων, ἔσχατον δὲ τῇ θέσει. τέλειος μὲν οὖν ὁ γίνεται συλλογισμὸς ὁδ' ἐν τούτῳ τῷ σχήματι, δυνατὸς δ' ἔσαι καὶ καθόλου καὶ μὴ καθόλου τῶν ὅρων ὄντων πρὸς τὸ μέσον. καθόλου μὲν οὖν ὄντων, ὅταν καὶ τὸ Π καὶ τὸ Ρ παντὶ τῷ Σ ὑπάρχῃ, ὅτι τινὶ τῷ Ρ τὸ Π ὑπάρξει ἐξ ἀνάγκης· ἐπεὶ γὰρ ἀντιστρέφει τὸ κατηγορικόν, ὑπάρξει τὸ Σ τινὶ τῷ Ρ, ὡς' ἐπεὶ τῷ μὲν Σ παντὶ τὸ Π, τῷ δὲ Ρ τινὶ τὸ Σ, ἀνάγκη τὸ Π τινὶ τῷ Ρ ὑπάρχειν· γίνεται γὰρ συλλογισμὸς διὰ τοῦ πρώτου σχήματος. ἔσι δὲ καὶ διὰ τοῦ ἀδυνάτου καὶ τῷ ἐκθέσθαι ποιεῖν τὴν ἀπόδειξιν· εἰ γὰρ ἄμφω παντὶ τῷ Σ ὑπάρχει, ἂν ληφθῇ τι τῶν Σ οἷον τὸ Ν, τούτῳ καὶ τὸ Π καὶ τὸ Ρ ὑπάρξει, ὥσε τινὶ τῷ Ρ τὸ Π ὑπάρξει. καὶ ἂν τὸ μὲν Ρ παντὶ τῷ Σ, τὸ δὲ Π μηδενὶ ὑπάρχῃ, ἔσαι συλλογισμὸς ὅτι τὸ Π τινὶ τῷ Ρ οὐχ ὑπάρξει ἐξ ἀνάγκης· ὁ γὰρ αὐτὸς τρόπος τῆς ἀποδείξεως ἀντιγραφείσης τῆς Ρ Σ προτάσεως. δειχθείη δ' ἂν καὶ διὰ τῆς ἀδυνάτου καθάπερ ἐπὶ τῶν πρότερον. ἐὰν δὲ τὸ μὲν Ρ μηδενὶ τὸ δὲ Π παντὶ ὑπάρχῃ τῷ Σ, οὐκ ἔσαι συλλογισμός. ὅροι τοῦ ὑπάρχειν ζῷον – ἵππος – ἄνθρωπος, τοῦ μὴ ὑπάρχειν ζῷον – ἄψυχον – ἄνθρωπος. ὁδ' ὅταν ἄμφω κατὰ μηδενὸς τοῦ Σ λέγηται, οὐκ ἔσαι συλλογισμός. ὅροι τοῦ ὑπάρχειν ζῷον – ἵππος – ἄψυχον, τοῦ μὴ ὑπάρχειν ἄνθρωπος – ἵππος – ἄψυχον· μέσον ἄψυχον. φανερὸν ἓν καὶ ἐν τούτῳ τῷ σχήματι πότ' ἔσαι καὶ πότ' οὐκ ἔσαι συλλογισμὸς καθόλου τῶν ὅρων ὄντων. ὅταν μὲν γὰρ ἀμφότεροι οἱ ὅροι ὦσι κατηγορικοί, ἔσαι συλλογισμὸς ὅτι τινὶ ὑπάρχει τὸ ἄκρον τῷ ἄκρῳ, ὅταν δὲ ζερητικοί, οὐκ ἔσαι. ὅταν δ' ὁ μὲν ᾖ ζερητικὸς ὁ δὲ καταφατικός, ἐὰν μὲν ὁ μεῖζων γένηται ζερητικὸς ἄτερος δὲ καταφατικός, ἔσαι συλλογισμός. ὅτι τινὶ οὐχ ὑπάρχει τὸ ἄκρον τῷ ἄκρῳ, ἐὰν δ' ἀνάπαλιν, οὐκ ἔσαι.

Ἐὰν δ' ὁ μὲν ᾖ καθόλου πρὸς τὸ μέσον ὁ δ' ἐν μέρει, κατηγορικῶν μὲν ὄντων ἀμφοῖν ἀνάγκη γίνεσθαι συλλογισμόν, ἂν ὁποτεροσῶν ᾖ καθόλου τῶν ὅρων. εἰ γὰρ τὸ μὲν Ρ παντὶ τῷ Σ τὸ δὲ Π τινί, ἀνάγκη τὸ Π τινὶ τῷ Ρ ὑπάρχειν. ἐπεὶ γὰρ ἀντιστρέφει τὸ καταφατικόν, ὑπάρξει τὸ Σ τινὶ τῷ Π, ὡς' ἐπεὶ τὸ μὲν Ρ παντὶ τῷ Σ, τὸ δὲ Σ τινὶ τῷ Π, καὶ τὸ Ρ τινὶ τῷ Π ὑπάρξει· ὥσε τὸ Π τινὶ τῷ Ρ. πάλιν εἰ τὸ μὲν Ρ τινὶ τῷ Σ τὸ δὲ Π παντὶ ὑπάρχει, ἀνάγκη τὸ Π τινὶ τῷ Ρ ὑπάρχειν· ὁ γὰρ αὐτὸς τρόπος τῆς ἀποδείξεως. ἔσι δ' ἀποδεῖξαι καὶ διὰ τοῦ ἀδυνάτου καὶ τῇ ἐκθέσει, καθάπερ ἐπὶ τῶν προτέρων. ἐὰν δ' ὁ μὲν ᾖ κατηγορικὸς ὁ δὲ ζερητικός, καθόλου δὲ ὁ κατηγορικός, ὅταν μὲν ὁ ἐλάττων ᾖ κατηγορικός, ἔσαι συλλογισμός. εἰ γὰρ τὸ Ρ παντὶ τῷ Σ, τὸ δὲ Π τινὶ μὴ ὑπάρχει, ἀνάγκη τὸ Π τινὶ τῷ Ρ μὴ ὑπάρχειν. εἰ γὰρ παντί, καὶ τὸ Ρ παντὶ τῷ Σ καὶ τὸ Π παντὶ τῷ Σ ὑπάρξει· ἀλλ' οὐχ ὑπῆρχεν. δείκνυται δὲ καὶ ἄνευ τῆς ἀπαγωγῆς, ἐὰν ληφθῇ τι τῶν Σ ᾧ τὸ Π μὴ ὑπάρχει. ὅταν δ' ὁ μεῖζων ᾖ κατηγορικός, οὐκ ἔσαι συλλογισμός, οἷον εἰ τὸ μὲν Π παντὶ τῷ Σ, τὸ δὲ Ρ τινὶ τῷ Σ μὴ ὑπάρχει. ὅροι τοῦ παντὶ ὑπάρχειν ἔμψυχον – ἄνθρωπος – ζῷον. τοῦ δὲ μηδενὶ οὐκ ἔσι λαβεῖν ὅρους, εἰ τινὶ μὲν ὑπάρχει τῷ Σ τὸ Ρ, τινὶ δὲ μή· εἰ γὰρ παντὶ τὸ Π τῷ Σ ὑπάρχει, τὸ δὲ Ρ τινὶ τῷ Σ, καὶ τὸ Π τινὶ τῷ Ρ ὑπάρξει· ὑπέκειτο δὲ μηδενὶ ὑπάρχειν. ἀλλ' ὥσπερ ἐν τοῖς πρότερον λεκτέον· ἀδιορίσῳ γὰρ ὄντος τοῦ τινὶ μὴ ὑπάρχειν καὶ τὸ μηδενὶ ὑπάρχον ἀληθὲς εἰπεῖν τινὶ μὴ ὑπάρχειν· μηδενὶ δὲ ὑπάρχοντος οὐκ ἦν συλλογισμός. φανερὸν ἓν ὅτι οὐκ ἔσαι συλλογισμός. ἐὰν δ' ὁ ζερητικὸς ᾖ καθόλου τῶν ὅρων, ὅταν μὲν ὁ μεῖζων ᾖ ζερητικὸς ὁ δὲ ἐλάττων κατηγορικός, ἔσαι συλλογισμός. εἰ γὰρ τὸ Π μηδενὶ τῷ Σ, τὸ δὲ Ρ τινὶ ὑπάρχει τῷ Σ, τὸ Π τινὶ τῷ Ρ οὐχ ὑπάρξει· πάλιν γὰρ ἔσαι τὸ πρῶτον σχῆμα τῆς Ρ Σ προτάσεως ἀντιγραφείσης. ὅταν δὲ ὁ ἐλάττων ᾖ ζερητικός, οὐκ ἔσαι συλλογισμός. ὅροι τοῦ ὑπάρχειν ζῷον – ἄνθρωπος – ἄγριον, τοῦ μὴ ὑπάρχειν ζῷον – ἐπιςήμη – ἄγριον· μέσον ἀμφοῖν τὸ ἄγριον. ὁδ' ὅταν ἀμφότεροι ζερητικοὶ τεθῶσιν, ᾖ δ' ὁ μὲν καθόλου ὁ δ' ἐν μέρει. ὅροι

3. ἐάν *C.* ‖ 4. ὅτι καὶ *C.* ‖ 9. ἀλλ' ἅπαντες *C.* ‖ 11. τοιοῦτο *B.* ‖ 17. μὲν ὄντων τῶν ὅρων *C.* ‖ 23. τῷ] τῶ corr *C.* ‖ 24. ἐὰν *C.* ‖ 25 et 26. ὑπάρχει *A.* ‖ 26. ἐὰν *C.* ‖ 28. τρόπος ἔσαι τῆς *C* ‖ 30. τῶν] τῶ *C.* ‖ μηδενὶ τὸ σ τὸ *C.* ‖ 38. τεθῶσι *C.*

2. ὁ ἕτερος *AC.* ‖ 8. σ ὑπάρχει, τὸ *C.* ‖ 11. ὥσε καὶ τὸ *C* et cort *B.* ‖ 15. ἐπὶ] καὶ ἐπὶ *C.* ‖ 17. σ ὑπάρχει, τὸ *C.* ‖ 20. ὑπῆρχε παντί. δείκνυται *C.* ‖ 22. κατηγορικός, ὁ δ' ἐλάττων μερικὸς ζερητικός, ἐκ rc *AB.* ‖ 28. ἀορίσῳ *A.* ‖ 30. ὑπάρχοντι pr *C.* ‖ οὐκ ἦν συλλογισμός om *C.* ‖ 31. ἓν om *AB.* ‖ 32. ὁ δὲ ἐλάττων κατηγορικὸς om *A.* ‖ 38. μέσον ἐν ἀμφοῖν *AB.* ‖ 39. ὅροι τοῦ μὴ ὑπάρχειν ὅταν *C* et rc *A.*

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Retained-file identities


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```
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```

fonts/literata-OFL.txt

```
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fonts/provenance.json


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]
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fonts/sourcesans3-OFL.txt

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